[INCH-POUND]
A-A-59927
13 January 2014

COMMERCIAL ITEM DESCRIPTION

PACKING MATERIAL, GRAPHITIC, WIRE REINFORCED

The General Services Administration has authorized the use of this Commercial Item Description, for all federal agencies.

1. Scope. This commercial item description covers requirements for a flexible, wire reinforced, graphitic packing. The packing is intended for steam or water applications and for temperatures up to 1200°F.

2. Salient characteristics. Salient characteristics shall be in accordance with 2.1 through 2.5. Preproduction testing for compliance with all characteristics that require testing (see 2.1.1, and 2.2 – 2.7) shall be performed on manufactured packing and shall be repeated every time there is a change in the raw materials, construction process or quality control process (see 4.2). Production testing shall be performed for each lot of packing (see 4.3). One production test sample is required. If tests are required, unless otherwise specified, the tests shall be conducted at ambient conditions.

2.1 Construction. The packing shall nominally consist of layers of graphite tape that are plied in compact strands. Each strand shall be reinforced with a Nickel-Chromium-Iron Alloy 600 per UNS N06600 (commonly known by its trademark as INCONEL® Alloy 600) wire mesh covering. For 0.25-inch cross section, the packing shall contain two center strands. The packing shall be manufactured using eight bobbins, with one strand per bobbin, and shall be square plait braided. The wire thickness of the Inconel wire mesh covering shall be 0.004 inch (nominal).

2.1.1 Detrimental materials. The packing shall be chemically analyzed using the methods in paragraphs 2.1.1.3 (b), (c), and (d). The detrimental materials shall not exceed the limits in Table 1.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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2.1.1.1 Prohibited additions. There shall be no intentional additions of any of the detrimental materials of Table 1, or of antimony, arsenic, bismuth, cadmium, lead, tin, or zinc during the manufacture or packaging of this product. The packing shall not contain lubrication agents or corrosion inhibitors.

2.1.1.2 Mercury exclusion. During manufacturing, fabrication, handling, packaging and packing, the packing material shall not come in contact with mercury or mercury-containing compounds.

2.1.1.3 Detrimental materials – Test procedure.

(a) Prior to the tests in (b), (c), and (d), the Inconel wires shall be removed from the packing.
(b) Total halogen content (consisting of fluorine, chlorine, and bromine) shall be determined by decomposing the product using Parr Bomb or Pyro hydrolysis techniques and analysis of the residue by ion chromatography, specific ion electrode, or spectrophotometric methods. Other equivalent methods may be used.
(c) Total sulfur content shall be determined by decomposing the product using Parr Bomb or Pyro hydrolysis techniques and analysis of the residue by gravimetric or turbidimetric methods. Other equivalent methods may be used.
(d) The concentration of mercury shall be determined by atomic absorption, plasma emission, emission spectrographic or X-ray fluorescence methods, as appropriate. Other equivalent methods may be used.

Table 1. Detrimental Materials

Element	Maximum allowable concentration (parts per million)
Mercury (Hg)	10
Sulfur (S)	750
Total halogens (Fluorine, Chlorine, and Bromine)	500
Fluorine (F)	250
Chlorine (Cl)	250
Bromine (Br)	250

2.1.1.4 Detrimental material and mercury exclusion certification. The results of the detrimental material analysis shall be submitted to the purchaser. A certification of detrimental material and mercury exclusion conformance shall be submitted to the purchaser.

2.1.1.5 Workmanship. The packing shall be free from extraneous material and visible defects that may affect its serviceability. Visual defects include but are not limited to holes or voids, rips or tears, creasing or crimping, delaminations, gouges, or rings not cleanly cut.

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2.2 Bulk density. The bulk density of the packing shall be greater than or equal to 72.5 pounds per cubic foot (lb/ft^3) [1.16 grams per cubic centimeter (g/cm^3)] and shall be tested in accordance with ASTM C 559.

2.3 Ash content. The ash content of the packing shall not exceed 1.5 percent and shall be tested in accordance with ASTM C 561. The mass of the Inconel wires shall not be included in the ash content calculation.

2.4 Carbon assay. The sample shall be dried to constant mass at 300±5°F before testing. The percent carbon shall be based upon mass of the dried sample. The carbon content shall be greater than or equal to 95 percent and shall be made in accordance with ASTM D 5373 or an alternate method of analysis with equal or improved accuracy and precision. The mass of the Inconel wires shall not be included in the carbon assay calculation.

2.5 Dimensions. The packing shall be supplied in pre-cut lengths and shall meet the dimensions of Naval Sea Systems Command (NAVSEA) drawing 820-8225659 when wrapped around a mandrel. NAVSEA Drawing 820-8225659 shall be provided to the vendor/supplier as part of the quote and order.

2.6 Visual examination. The packing shall be visually examined for surface defects and conformance with NAVSEA drawing 820-8225659. Packing containing one or more visual defects that may affect serviceability shall be rejected.

2.7 Pressure test. The supplier shall provide twenty (20) rings per NAVSEA Drawing 820-8225659 to Manager, Technical Services, Bechtel Plant Machinery, Inc., 3500 Technology Drive, Monroeville, PA 15146 (BPMI). BPMI will pressure test these rings, consisting of a packing leakage test and a cyclic test, for satisfactory leak performance at a facility of BPMI's choosing. Additionally, BPMI will perform a throttle test of the packing using prototypic conditions at a facility of BPMI's choosing.

3. Regulatory requirements. All appropriate regulatory requirements that apply to items offered on the commercial market shall apply to those offered to the Government.

4. Quality assurance provisions.

4.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description in accordance with the required frequency of testing specified in 2, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be provided for under the provisions of the contract.

4.2 Preproduction requirements. Prior to supplying material to this CID, the supplier shall provide objective evidence of successful completion of preproduction testing, consisting of 2.1.1, and 2.2 – 2.7. Preproduction testing shall be

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repeated if there is a change in raw materials, construction process, or quality control process.

4.3 Production requirements. The supplier shall certify compliance with 2.1.1 (detrimental materials and mercury exclusion) and shall provide objective evidence of compliance with 2.2 (bulk density), 2.3 (ash content), 2.4 (carbon assay), 2.5 (dimensions), and 2.6 (visual examination) in accordance with the frequency of testing specified in 2.

5. Preservation packaging, labeling and marking. The ends of the packing shall be taped. The tape shall leave no visible residue on the packing once the tape is removed. The packing shall be unit packed. Preservation, labeling, marking and other packaging requirements shall be as specified in the contract or order, except as a minimum, the following marking shall appear on the innermost container and all other containers or packaging in prominent letters:
(a) Packing material per CID A-A-59927
(b) Store in a cool, dry place
(c) Date of manufacture
(d) Manufacturer's instructions for use

6. Notes.

6.1 Suggested source of supply.
Chesterton MPCS 1601-NCI
MPCS: Mechanical Packing Custom Solution, NCI: No Corrosion Inhibitor
(Manufacturer: A. W. Chesterton, 860 Salem St., Groveland, MA 01834)

6.2 Sources of reference documents.

6.2.1 Copies of ASTM Standards are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

6.3 Keywords.

Packing, Graphitic

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